

(Cs601)

Data communication
Final term notes
Topic (124 - 220)

Topic 124

second layer of OSI
Data link layer:

The internet is a combination of networks glued together by connecting device (router, switches). If a packet is to travel from a host to another host, it needs to pass through these networks. Data link layer control node-to-node communication.

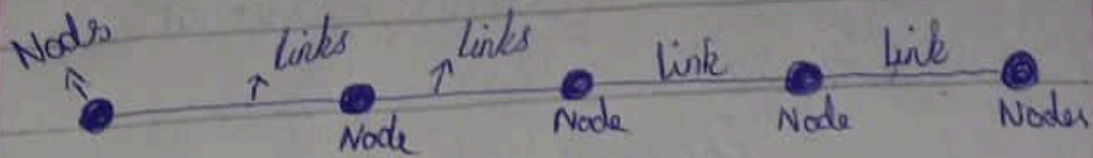
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Topic 125

Nodes & Links:

Communication in data link is node-to-node. Router are nodes and network

is between as link.



Topic 126

Data link layer services:

Data link

provide four services:

- Framing
- Flow control
- Error control
- Congestion control

Framing:

Data link layer divides data packets into small frames.

Framing is a primary service of data link layer. Data link layer needs to pack bits into frames, so that each frame is distinguishable from another.

Topic 128

Types of data link layer:

There are

two type of dll.

- Point-to-point link
- Broadcast link

Point-to-point:

It refers to the communication connection between two communication endpoints or nodes e.g: telephone call.

Broadcast link:

It connects two or more nodes and support broadcast transmission, where one node can transmit so that all nodes can receive the same transmission. e.g: Ethernet.

Sub-layer of DLL:

There are

two sublayer of DLL are given below:

- DLC (Data link control)
- MAC (Media access control)

DLC:

Data link control deals with point-to-point link as well as broadcast link.

MAC:

Media access control deals with only broadcast link.

Topic 129-130

IP address:

IP address are the identifiers at the network layer in internet we cannot make a packet reach its destination using only IP address.

Link layer addresses:

Source and

destination addresses two ends but cannot define which links the packet and path will take. Encapsulation and decapsulation process are involved. It is also known as physical or MAC addresses.

Topic 131

Type of addressing:

There are three types of addressing:

- Unicast
- Multicast
- Broadcast

Unicast:

One to one communication.

Used to describe communication where piece of information is sent from one point to another

point (one sender & one receiver).

Multicast:

Term used to describe communication where piece of information is sent from one or more points to a set of other points. (more sender & more receiver).

Broadcast:

Term: used to describe communication where a piece of information is sent from one point to all other points. (only one sender & multi receiver:).